# Ultra-High-Speed DWDM Photonic Transport Link – Investigative Report (ECA18 CO4)

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## Executive Summary

The recommended design is a 32-channel DP-16QAM coherent DWDM link at 31.25 GBd on a 50 GHz grid (6.4 Tb/s net, 4 b/s/Hz), built on silicon-photonic IQ Mach-Zehnder modulators, hybrid distributed-Raman + EDFA amplification over 80 km spans, and a coherent DSP receiver without digital backpropagation in the baseline. Analytical (GN-model) evaluation gives about 3.6 dB OSNR/SNR margin over the pre-FEC threshold (BER $2\times10^{-2}$) after 10 spans (800 km). The final decision is **YES – deploy**, with conditions listed at the end.

**Assumptions used throughout** (the assignment leaves some values blank, so these are stated explicitly): SMF-28, 80 km spans, $\alpha$ = 0.2 dB/km, D = 16.8 ps/(nm·km), $\gamma$ = 1.3 $W^{-1}km^{-1}$, PMD coefficient 0.1 ps/√km, span loss 18 dB (80 km fibre plus connectors/splices), 25 % overhead soft-decision FEC with pre-FEC BER threshold $2\times10^{-2}$, C-band 1550 nm (193.4 THz).

**Honesty note on simulation:** OptiSystem/OptSim/MATLAB were not available in this session. All performance numbers below come from closed-form equations and a Gaussian-noise (GN) model coded in Python. The simulation setup in Stage 6 gives the exact component settings so you can reproduce the schematics, eye diagrams, constellations and spectra in OptiSystem and paste your own screenshots; the analytical values are what those simulations should approximately reproduce.

## Stage 1 – Interpret the Communication Requirement (15 marks)

The service needs 32 × 200 Gb/s = 6.4 Tb/s of aggregate payload capacity, which a 50 GHz-grid C-band system delivers only with a coherent, polarization-multiplexed higher-order format.

### 1.1 Literature and standards review

| Source | Relevance to this design |
|---|---|
| ITU-T G.694.1 (DWDM frequency grid) and G.698.x (multichannel DWDM application codes) | Fixes the 12.5/25/50/100 GHz anchor grid around 193.1 THz and the channel-power, OSNR and dispersion limits an optical interface must respect; the 50 GHz grid is used here. |
| IEEE 802.3ct (100GBASE-ZR, DP-QPSK, 80 km DWDM) | Establishes coherent DWDM with soft-decision FEC as the accepted Ethernet route for 80 km amplified links. |
| Source | Relevance to this design |

| | |
|---|---|
| IEEE 802.3ck (100/200/400 Gb/s per lane electrical interfaces, 106.25 GBd PAM4) | Defines the host-side electrical interface that feeds the optical engine; PAM4 at about 106 GBd is the electrical reference. |
| OIF-CEI-112G and OIF 400ZR/ZR+ | Benchmark of a deployed DP-16QAM coherent pluggable (about 60 GBd, 75 GHz spacing). It shows DP-16QAM is commercially mature and fixes DSP power targets (about 15–20 W per 400G module). |
| Coherent transmission literature (Kikuchi 2016; Savory 2010; Poggiolini GN model 2014; Ip & Kahn digital backpropagation 2008; Essiambre et al. 2010 capacity limits) | Provides the DSP chain, the nonlinear noise model and the capacity-limit framing used in Stages 2–5. |
| Silicon-photonic and EO-polymer modulator literature (Reed et al. 2010; Witzens 2018; Lauermann/Koos et al. for SOH; Alloatti et al.) | Provides bandwidth, VπL and loss values for the modulator comparison. |

### 1.2 Aggregate capacity and spectral requirement

$$C_{agg} = N_{ch} \times R_{ch} = 32 \times 200\ \text{Gb/s} = 6.4\ \text{Tb/s}$$

On a 50 GHz grid the occupied optical bandwidth is 32 × 50 GHz = 1.6 THz, about 12.8 nm, which fits inside the C-band (about 4.4 THz, 35 nm). The net spectral efficiency is 200 Gb/s ÷ 50 GHz = **4 b/s/Hz**. The line rate including 25 % FEC overhead is 250 Gb/s per channel, or 8 Tb/s for the whole link.

### 1.3 Physical-layer metrics to investigate

| Metric | Why it matters | Baseline target |
|---|---|---|
| Spectral efficiency | Sets channel count per fibre | ≥ 4 b/s/Hz net |
| Required OSNR (0.1 nm) | Sets amplifier noise budget | about 16.7 dB theory (SNR 12.7 dB + 10log(31.25/12.5)); about 18.5 dB with implementation penalty, for BER $2\times10^{-2}$ |
| Pre-FEC BER / Q-factor | Pass/fail against SD-FEC | BER ≤ $2\times10^{-2}$ (Q ≥ 6.25 dB) |
| Launch power per channel | Trade-off between ASE and Kerr noise | about 0 dBm optimum (derived in Stage 3) |
| Modulator 3 dB bandwidth | Limits baud rate and ISI | ≥ 25 GHz electro-optic |
| Insertion loss of PIC, MMI, AWG | Reduces launch power, forces extra gain | ≤ 6 dB modulator, ≤ 5 dB AWG mux |
| Metric | Why it matters | Baseline target |

| | | |
|---|---|---|
| Power per bit | Operating cost | ≤ 100 pJ/b end to end (about 20 W per 200G transceiver) |
| Footprint | Integration feasibility | single PIC per channel, pluggable class |

### 1.4 Impairment-to-DSP mapping

| Impairment | Origin | Matching compensation |
|---|---|---|
| Chromatic dispersion | Group-velocity dispersion, D = 16.8 ps/(nm·km); 800 km accumulates about 13 440 ps/nm | Static frequency-domain equalizer (FDE) |
| PMD and polarization rotation | Random birefringence; mean DGD = 0.1 × √800 = 2.8 ps | Adaptive 2×2 MIMO (CMA/RDE then decision-directed LMS) |
| SPM, XPM | Kerr effect, intensity-dependent phase | Optimized launch power, optionally DBP |
| FWM | Mixing of equal-spaced channels, strongest at low local dispersion | Dispersive fibre (SMF-28) suppresses it; unequal-grid not needed |
| ASE noise | Optical amplifiers | Low-NF amplifier (Raman), not removable by DSP |
| Laser phase noise, frequency offset | Free-running Tx and LO lasers | Frequency-offset estimator + carrier phase recovery |
| AWG/MMI crosstalk and filtering | Finite isolation (typically ≥ 25 dB) and passband narrowing | Per-channel guard band, adaptive equalizer taps |
| PDL | Polarization-dependent loss of components | MIMO equalizer (partial), component specification |

### 1.5 Practical trade-offs

Silicon photonics offers CMOS-compatible wafer-scale production, compact (mm-scale) devices and co-integration of drivers, but its depletion-mode modulators have about 4–6 dB insertion loss and need higher drive swing. Thin-film lithium niobate offers the lowest loss and highest bandwidth (above 70 GHz) but is a less mature integration platform. Electro-optic polymers (including silicon-organic hybrid devices) offer very low $V\pi$ (below 1 V) but raise reliability and thermal-stability questions. A DP-16QAM transceiver also raises DSP power, so the DSP node and the amplifier plan must be chosen together.

### 1.6 Investigation Requirement Statement

> The DWDM link shall carry 32 channels of 200 Gb/s net (6.4 Tb/s) on the ITU-T 50 GHz C-band grid at ≥ 4 b/s/Hz net spectral efficiency, with a pre-FEC BER ≤ $2\times10^{-2}$ (Q ≥ 6.25 dB) and at least 3 dB OSNR margin at end of life after 800 km (10 × 80 km SMF-28 spans), using integrated photonics (modulator loss ≤ 6 dB, bandwidth ≥ 25 GHz), per-channel launch power near the nonlinear optimum (about 0 dBm), and a transceiver power budget of about 20 W per 200G channel, while remaining consistent with ITU-T G.694.1/G.698, IEEE 802.3ct/ck and OIF-CEI practice.

## Stage 2 – Modulation Format and Modulator Selection (20 marks)

DP-16QAM at 31.25 GBd is the only candidate that delivers 200 Gb/s inside a 50 GHz slot while keeping the required OSNR below 17 dB and the modulator bandwidth near 25 GHz.

## 2.1 Baud rate and channel spacing calculation

The 200 Gb/s net rate is raised by the 25 % soft-decision FEC overhead to a 250 Gb/s line rate. For a format carrying *m* bits per symbol per polarization with *P* polarizations:

```
R_s = \frac{R_{line}}{m \cdot P}, \qquad B_{occ} \approx R_s (1+\beta), \qquad SE = \frac{R_{net}}{\Delta f}
```

- **DP-16QAM:** m = 4, P = 2, so R_s = 250 / 8 = **31.25 GBd**. With a root-raised-cosine roll-off β = 0.1 the occupied bandwidth is 34.4 GHz, which fits a **50 GHz** slot with 15.6 GHz guard band. SE = 200/50 = **4 b/s/Hz**.
- **PAM4 (IM-DD, single polarization):** m = 2, P = 1. IEEE 802.3ck-style RS(544,514) FEC adds 5.8 %, giving 211.7 Gb/s and R_s = **105.8 GBd** (the standardized lane rate is 106.25 GBd). The double-sideband signal occupies about 106 GHz, so it needs a **100 GHz** slot and gives SE = 2 b/s/Hz, half of DP-16QAM. A 106 GBd signal also needs about 60–70 GHz electrical bandwidth, which is beyond today's silicon modulators without heavy equalization.

### 2.2 Candidate format comparison

Required SNR is the theoretical per-symbol SNR at the pre-FEC BER of $2\times10^{-2}$ (Gray-coded square QAM: BER ≈ $(4/m)(1 - 1/\sqrt{M}) \cdot (1/2)\,\mathrm{erfc}(\sqrt{(3\,\mathrm{SNR}/(2(M-1)))})$, and required OSNR = SNR + $10\log_{10}(R_s / 12.5\ \mathrm{GHz})$.

| Format | Bits/symbol (total) | Baud for 200G net (GBd) | Slot (GHz) | Net SE (b/s/Hz) | Required SNR at BER $2\times10^{-2}$ (dB) | Required OSNR 0.1 nm (dB) | Impairment tolerance |
|---|---|---|---|---|---|---|---|
| NRZ-OOK (IM-DD) | 1 | about 250 | 300 | 0.7 | high (direct detection, ~ 15–17 Q-dB | > 30 | CD limits reach to a few km at this rate |

| Format | Bits/symbol (total) | Baud for 200G net (GBd) | Slot (GHz) | Net SE (b/s/Hz) | Required SNR at BER $2\times10^{-2}$ (dB) | Required OSNR 0.1 nm (dB) | Impairment tolerance |
|---|---|---|---|---|---|---|---|
| | | | | | class) | | |
| DPSK (balanced detection) | 1 | about 250 | 300 | 0.7 | about 3 dB better than OOK | about 25 | CD and nonlinear phase noise |
| DQPSK | 2 | about 125 | 150 | 1.3 | about 11 (differential detection) | about 25 | Moderate CD tolerance, no electronic polarization recovery |
| PAM4 (IM-DD) | 2 | 105.8 | 100 | 2.0 | about 14.6 electrical SNR | > 24 | Severe CD penalty at 106 GBd without DSP; no phase/polarization info |
| DP-QPSK | 4 | 62.5 | 75 | 2.7 | 6.25 | 6.25 + 10log(62.5/12.5) = 13.2 | Excellent; used in 100G ZR |
| **DP-16QAM** | **8** | **31.25** | **50** | **4.0** | **12.7** | **16.7** | Good with coherent DSP; needs low-noise links |
| DP-64QAM | 12 | 20.8 | 37.5 | 5.3 | 18.4 | 20.9 | Needs about 5.7 dB more SNR than 16QAM; reach falls to about one quarter |

DP-QPSK has the lowest OSNR requirement but needs a 75 GHz slot (2.7 b/s/Hz), so it cannot reach the 6.4 Tb/s-in-1.6 THz goal on the 50 GHz grid. DP-64QAM would exceed the achievable transceiver SNR (about 17–18 dB, limited by DAC/ADC ENOB and driver noise). DP-16QAM is therefore the balanced optimum.

### 2.3 Modulation mechanisms

Typical literature values; the $V\pi\cdot L$ product is the length-weighted voltage needed for a $\pi$ phase shift.

| Mechanism | Physical effect | Speed limit | Vπ / strength | Chirp | Verdict |
|---|---|---|---|---|---|
| Electro-optic (Pockels, $LiNbO_3$ / TFLN / EO polymer) | Field-induced refractive-index change, $\Delta n = \frac{1}{2} n^3 r E$ | > 100 GHz (fs-scale intrinsic) | LN about 2–3 V·cm (TFLN), polymer SOH about 0.05–0.1 V·cm | Zero in push-pull | Suitable |
| Free-carrier plasma dispersion (silicon, carrier depletion) | Carrier density changes n and absorption | 40–60 GHz | about 1.5–2.5 V·cm | Small, mixed amplitude/phase | **Suitable, CMOS-compatible** |
| Acousto-optic | Bragg diffraction from an acoustic wave | Below about 1 GHz (acoustic transit) | Tens of mW RF | Frequency shift | Unsuitable for 30 GBd data; useful for tunable filters |
| Thermo-optic | Temperature-dependent index ($dn/dT$ about $1.8\times10^{-4}$ /K in Si) | 10 kHz–100 kHz (µs–ms) | about 20 mW per π | None | Only for bias and phase trimming |
| Magneto-optic (Faraday) | Magnetic-field-induced polarization rotation | MHz-class | Non-reciprocal | None | Isolators and circulators, not data modulation |
| Kerr ($\chi^3$) | Intensity-dependent index, $n_2$ | fs response | Needs very high intensity | Strong SPM | All-optical switching only; here it is the impairment, not the tool |

For the Pockels effect in a Mach-Zehnder, the switching voltage is:

```
V_\pi = \frac{\lambda\, g}{2\, n^{3} r_{33}\, \Gamma\, L}
```

The MZM output field is $E_{out} = E_{in} \cos(\pi V/2V\pi)$ (push-pull), so a 16QAM I/Q driver should stay inside about ±0.35 Vπ to limit nonlinearity.

### 2.4 MZI versus micro-ring resonator, silicon photonics versus EO polymer

| Property | Silicon-photonic MZM (travelling wave) | Silicon micro-ring modulator (MRR) | EO polymer / SOH MZM |
|---|---|---|---|
| 3 dB electro-optic bandwidth | 30–50 GHz | 30–50 GHz, but limited by photon lifetime (Q about 5000–10 000) | 40–100 GHz |
| Footprint | 3–4 mm arms, about 0.5 mm² per IQ | about 10 µm radius, about 100 µm² | 0.5–2 mm |
| Insertion loss | 4–6 dB | 1–3 dB | 4–8 dB (slot-waveguide loss) |
| Thermal sensitivity | Low: broadband, bias drift only | **High: about 80 pm/K, so about 10 GHz/K resonance shift** | Moderate; polymer poling stability limits temperature |
| Transfer function | Linear near quadrature, good for 16QAM | Lorentzian, nonlinear and bandwidth/extinction trade-off | Linear |
| Drive voltage | Vπ about 4–6 V for 3 mm | 1–2 V | Vπ below 1 V |
| Wavelength dependence | Wideband, all C-band channels use one design | Narrowband, needs per-channel heater and lock | Wideband |
| Maturity | High (foundry PDKs, coherent modules shipping) | Medium (WDM-comb links) | Low to medium |

The MRR is excellent for compact, low-energy intensity modulation and WDM comb links, but 16QAM needs a linear, thermally robust IQ transfer function. A ring of Q about 8000 at 1550 nm has a photon lifetime of about 6.6 ps and an optical linewidth near 24 GHz, which already limits a 31 GBd signal. The silicon MZM, with a foundry-supported dual-polarization IQ layout, is the safer choice.

Example drive calculation: for a 3 mm silicon arm with VπL = 1.8 V·cm, Vπ = 1.8 / 0.3 = **6 V**. Keeping the drive at 0.35 Vπ per arm in push-pull needs about 2.1 V peak per arm, roughly 3–4 V peak-to-peak from a SiGe driver, which is within reach of current driver ICs. Thin-film lithium niobate (Vπ about 2 V, bandwidth above 60 GHz) is the identified upgrade path if driver power must be cut.

### DESIGN DECISION 1 – Modulation / modulator configuration

**I recommend DP-16QAM at 31.25 GBd (net 200 Gb/s, 250 Gb/s line, 50 GHz slot) generated by a carrier-depletion silicon-photonic dual-polarization IQ Mach-Zehnder modulator (push-pull, travelling-wave electrodes, about 3 mm arms, 3 dB bandwidth ≥ 30 GHz, Vπ about 6 V) because** it is the only candidate that reaches 4 b/s/Hz on the 50 GHz grid (PAM4 reaches 2 b/s/Hz at 106 GBd, DP-QPSK 2.7 b/s/Hz), its required OSNR of 16.7 dB (theory) is reachable with Raman-assisted amplification over 800 km, the MZM transfer function is linear and thermally robust where a micro-ring is not, and the silicon

platform co-integrates the MMI splitters, polarization beam combiner and heaters in a CMOS-compatible, compact PIC.

## Stage 3 – Optical Amplification Configuration (20 marks)

A hybrid backward-pumped distributed-Raman plus EDFA span doubles the reach of an EDFA-only link (about 3200 km versus 1600 km at 1 dB margin) because it lowers the effective span noise figure from about 5 dB to about 0 dB.

### 3.1 Amplifier technology comparison

Typical literature values for a 32-channel C-band system.

| Amplifier | Gain bandwidth | Saturated output power | Noise figure | Gain tilt / ripple | Multichannel crosstalk | Verdict |
|---|---|---|---|---|---|---|
| EDFA | 32–35 nm (C-band) | +20 to +23 dBm | 4.5–5.5 dB | ±3 dB raw; ±0.5 dB with GFF | Negligible (ms gain dynamics) | Workhorse, used as booster/pre-amp |
| SOA | 30–60 nm | +10 to +15 dBm | 7–9 dB | Polarization- and wavelength-dependent | Severe: ns gain dynamics cause XGM, XPM and FWM between channels | Unsuitable for 32-channel WDM trunk |
| Lumped Raman | Set by pump wavelengths (up to 100 nm) | Pump-limited, needs 1–2 W | 6–8 dB (short gain fibre) | Pump-spacing dependent | Pump-signal and signal-signal Raman tilt | Niche (extra bandwidth) |
| Distributed Raman (backward pumped) | Set by pumps, flat with 2–3 pumps | Gain is in the transmission fibre | **About −1 to 0 dB effective** | ±0.5 dB with 2–3 pumps | Low; backward pumping averages RIN | Excellent noise improvement |
| **Hybrid Raman + EDFA** | C-band or wider | EDFA-limited | **About 0 dB effective span NF** | ±0.5 dB after GFF/DGE | Low | **Selected** |

### 3.2 Analytical problem: total output power and span gain

The assignment leaves the per-channel values blank, so the following values are assumed: input **−18 dBm per channel** (end of an 80 km span with 18 dB loss and 0 dBm launch) and target output **0 dBm per channel**, for N = 32 channels.

$$P_{out,tot}(\text{dBm}) = P_{ch,out} + 10\log_{10} N = 0 + 10\log_{10} 32 = 15.05\ \text{dBm}$$

$$P_{out,tot} = 10^{1.505}\ \text{mW} = 32.0\ \text{mW} = 0.032\ \text{W}$$

$$G_{span} = P_{ch,out} - P_{ch,in} = 0 - (-18) = 18\ \text{dB}\ (\times 63.1)$$

The total input is −3 dBm (0.51 mW). A booster or in-line EDFA with +17 dBm saturation power therefore runs with about 2 dB of output-power headroom, which keeps all 32 channels unsaturated.

General form for any assumed inputs: $P_{out,tot} = P_{ch,out} + 15.05$ dBm and $G = P_{ch,out} - P_{ch,in}$.

### 3.3 Gain flatness, GFF and dynamic equalization

A raw EDFA spectrum varies by about 6 dB (±3 dB) across 1530–1565 nm. Without correction the weakest channel in a cascade loses OSNR by the full excess gain difference, and ripple accumulates up to N times in a cascade of N identical amplifiers (10 spans × 0.5 dB residual ripple = up to 5 dB spread in the worst case). The design therefore uses:

- a fixed gain-flattening filter (thin-film or long-period grating) inside every EDFA to reach ±0.5 dB,
- two-pump Raman gain (1425 nm and 1455 nm pumps) whose tilt is complementary to the EDFA tilt,
- a wavelength-selective switch (dynamic gain equalizer) every 4 spans to re-level channel powers to ±0.3 dB, and per-channel pre-emphasis at the transmitter.

### 3.4 Cumulative ASE and OSNR over cascaded spans

For N identical spans of loss L (dB) and effective noise figure NF (dB), with 0.1 nm reference bandwidth at 1550 nm:

$$OSNR_{0.1nm}\ (\text{dB}) \approx 58 + P_{ch}(\text{dBm}) - L - NF - 10\log_{10} N$$

For 10 spans, L = 18 dB: an EDFA-only link at 0 dBm gives 58 + 0 − 18 − 5 − 10 = **25.0 dB**; the hybrid link at −1.7 dBm with NF about 0 dB gives 58 − 1.7 − 18 − 0 − 10 = **28.3 dB** (ASE only). The nonlinear-interference (NLI) term then adds a cubic noise that grows with launch power, so the best launch power is where ASE and NLI noise are balanced (optimum near 0 dBm for EDFA-only and near −1.7 dBm for hybrid, see the table in Stage 6).

The table below gives total SNR margin over the 12.7 dB theoretical threshold (BER $2\times10^{-2}$) at each link's optimum launch power, including a 17.5 dB transceiver-SNR ceiling:

| Spans (distance) | EDFA-only margin (dB) | Hybrid Raman + EDFA margin (dB) |
|---|---|---|
| 8 (640 km) | 2.9 | 3.8 |
| **10 (800 km), design point** | **2.5** | **3.6** |
| Spans (distance) | EDFA-only margin (dB) | Hybrid Raman + EDFA margin (dB) |
| 12 (960 km) | 2.2 | 3.4 |
| 16 (1280 km) | 1.6 | 3.0 |
| 20 (1600 km) | 1.0 | 2.7 |
| 24 (1920 km) | 0.6 | 2.3 |
| 32 (2560 km) | −0.3 | 1.7 |
| 40 (3200 km) | −1.0 | 1.2 |

### 3.5 Hybrid span power budget (design values)

| Point in span | Per-channel power (dBm) | Total, 32 ch (dBm) |
|---|---|---|
| Fibre launch (after booster) | −1.5 | 13.5 (22.6 mW) |
| After 80 km fibre (18 dB loss) | −19.5 | −4.5 |
| After distributed Raman, 10 dB on-off gain (backward pumps, about 0.9 W total) | −9.5 | 5.5 |
| After in-line EDFA, 8 dB gain, NF 5 dB | −1.5 | 13.5 |

**DESIGN DECISION 2 – Amplification configuration**

**I recommend a hybrid amplifier: a backward-pumped two-pump distributed Raman stage (about 10 dB on-off gain) followed by a gain-flattened EDFA (8 dB gain, 17 dBm saturated output, NF 5 dB) in every 80 km span, with a wavelength-selective-switch gain equalizer every 4 spans, because** the link needs 18 dB span gain and 13.5 dBm total output (22.6 mW) that an EDFA alone can supply, but only the Raman stage lowers the effective span NF to about 0 dB, which gives 28.3 dB ASE-limited OSNR after 10 spans (versus 25 dB) and about 3.6 dB end-of-link margin (versus 2.5 dB). SOAs are rejected because their ns gain dynamics produce inter-channel crosstalk and FWM in a 32-channel link; lumped Raman is rejected because it needs 1–2 W of pump for a worse noise figure.

**Stage 4 – WDM Impairment and DSP Compensation (20 marks)**

All linear impairments (CD, PMD, polarization rotation, laser phase noise) are best removed in a coherent DSP receiver with no dispersion-compensating fibre, and digital backpropagation is

kept as an
optional upgrade because its gain on a fully loaded 32-channel link is small compared with its power cost.

### 4.1 Physical mechanisms

| Impairment | Mechanism | Magnitude in this link (800 km, 31.25 GBd) |
|---|---|---|
| Chromatic dispersion (CD) | Wavelength-dependent group velocity, $\beta_2 = -D\lambda^2/(2\pi c) \approx$ | Accumulated 16.8 × 800 = 13 440 ps/nm; pulse spread ≈ 13 440 × 0.25 nm ≈ 3.4 ns ≈ 105 symbol |
| Impairment | Mechanism | Magnitude in this link (800 km, 31.25 GBd) |
| | $-21.4$ ps²/km | periods |
| Polarization-mode dispersion (PMD) | Random birefringence splits the two polarization modes in time | Mean DGD = 0.1 × √800 = 2.8 ps ≈ 0.09 symbol; time-varying, kHz-scale tracking needed |
| SPM | Intensity-dependent phase of a channel from its own power, $\varphi = \gamma P L_{eff}$ | At 0 dBm, $L_{eff}$ = 21.6 km per span gives $\gamma P L_{eff}$ = 0.028 rad per span |
| XPM | Phase shift induced by the intensity of neighbouring channels; dominant inter-channel effect | Averaged down by walk-off (neighbouring channels 50 GHz apart drift 16.8 × 0.4 nm ≈ 6.7 ps/km, about 0.2 symbol per km, or about 17 symbols per 80 km span) |
| FWM | Mixing of three channels creates a fourth at $f_i + f_j - f_k$ | Strongly suppressed by the high local dispersion of SMF-28 (phase mismatch), typically 30 dB or more below the signal at this launch power |
| PDL | Polarization-dependent loss in couplers, isolators and WSS | Typically 0.1–0.3 dB per element; accumulates as a statistical SNR penalty |
| ASE, transceiver noise | Amplifiers, DAC/ADC, drivers | Handled by Stage 3 and the transceiver SNR of 17.5 dB |

### 4.2 Optical versus digital compensation

| Approach | Strengths | Weaknesses | Decision |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Dispersion-compensating fibre (DCF) | Passive, no power | 5–10 dB extra loss per module, small core ($\gamma$ about 5 $W^{-1}km^{-1}$) gives stronger nonlinearity, added NF and latency, fixed amount; and a fully compensated link has *worse* nonlinear tolerance (channels stay aligned, so XPM/FWM do not average out) | Rejected |
| Optical phase conjugation (mid-span) | Compensates CD and some SPM/XPM for the whole band | Needs a nonlinear device with tens of dB gain, one symmetric midpoint, very hard to equalize across 32 channels, no field maturity | Rejected |
| Optical PMD compensator | Reduces DGD | Slow, narrowband, per channel hardware | Rejected |
| Digital coherent | Compensates CD, PMD, phase noise and | Costs DSP power and ADC/DAC resolution | **Selected** |
| Approach | Strengths | Weaknesses | Decision |
| DSP | frequency offset adaptively, tracks changes in microseconds, no extra optical loss, flexible reach | | |

### 4.3 Coherent DSP blocks

**Static CD equalization.** CD is a pure all-pass phase filter $H(\omega) = \exp(j\ \beta_2\ \omega^2\ L\ /\ 2)$. Implemented in the frequency domain (overlap-save FFT, 1024-point blocks) it needs far fewer operations than a time-domain FIR. The time-domain equivalent tap count for 2 samples per symbol ($F_s = 62.5$ GS/s) is:

```
N_{taps} = 2\left\lfloor \frac{|D|\, \lambda^{2} L F_s^{2}}{2c} \right\rfloor + 1
\approx 2\lfloor 210 \rfloor + 1 = 421
```

**Adaptive polarization demultiplexing and PMD compensation.** A 2×2 MIMO butterfly of four complex FIR filters (21 taps each at 2 samples/symbol, spanning about 10 symbols, far more than the 2.8 ps DGD plus residual filtering ISI require) inverts the unknown Jones matrix. For 16QAM the filters are pre-converged with the constant-modulus algorithm (CMA) and then refined with a radius-directed equalizer (RDE) and finally decision-directed LMS. CMA minimizes $J = E[(|y|^2 - R)^2]$, is blind and phase-insensitive, so it converges before carrier recovery.

**Digital backpropagation (DBP).** DBP numerically propagates the received field backwards through a virtual fibre with opposite sign of $\beta_2$ and $\gamma$ using the split-step Fourier method:

alternating a linear step (the CD filter, per step) and a nonlinear step (phase rotation $\varphi = -\gamma |E|^2 \Delta z$). A practical design uses 1–2 steps per span, so 10–20 CD-filter pairs for 800 km.

| DBP option | Typical SNR gain on a 32-channel link | Relative complexity | Verdict |
|---|---|---|---|
| None (baseline) | 0 dB | 1× | Selected for the baseline |
| Single-channel DBP, 1 step/span | About 0.3–0.5 dB (XPM from 31 neighbours remains) | About 10× the CD block, +30–60 % DSP power | Optional upgrade for reach > 2000 km |
| Full-field DBP (all 32 channels received together) | Up to 3–5 dB in literature | Not realizable: needs 1.6 THz receiver bandwidth and phase-locked LOs | Rejected |

### 4.4 Receiver DSP pipeline and why the order matters

1. **ADC and front-end correction** (62.5 GS/s, ENOB about 5.5 bits): IQ skew and amplitude/orthogonality imbalance (Gram-Schmidt) are fixed first, because every later block assumes clean I/Q.
2. **Static CD compensation (FDE, 421-tap equivalent):** CD is large (about 105 symbols), deterministic and common to both polarizations, so it must come first; the adaptive MIMO filters (21 taps) could not span it.
3. **Timing recovery** (Gardner or Godard, plus interpolation): needed so the 2 samples per symbol feed the equalizer aligned; clock offset is tracked at kHz bandwidth.
4. **Adaptive MIMO polarization demultiplexing/PMD compensation** (CMA → RDE → DD-LMS): works on the symbol-rate stream before carrier recovery because CMA is blind to phase, so it needs neither frequency offset nor phase correction.
5. **Frequency offset estimation** (fourth-power spectral line or pilot-aided, ±3 GHz range): removes the Tx/LO laser difference; the phase estimator can only track a few MHz of residual offset.
6. **Carrier phase estimation** (blind phase search, 32 test angles, 32-symbol blocks, with cycle-slip correction): with a combined 200 kHz linewidth the product $\Delta\nu \cdot T_s = 6.4\times10^{-6}$, well inside the 16QAM tolerance of about $10^{-4}$.
7. **Symbol demapping and soft-decision FEC decoding:** log-likelihood ratios are fed to the 25 % overhead SD-FEC; pre-FEC BER up to $2\times10^{-2}$ is corrected to below $10^{-15}$.

### 4.5 Expected improvement

Without CD compensation the eye is closed (about 105 symbols of ISI, BER near 0.5). With the full pipeline the receiver reaches the transceiver-limited SNR; the combined link SNR at the optimum launch power is 16.3 dB (hybrid, 10 spans), giving pre-FEC BER of about $1.3\times10^{-3}$ and Q about 9.6 dB, compared with the $2\times10^{-2}$ threshold (Q 6.25 dB). Adding single-channel DBP would raise Q by about 0.3–0.5 dB, which is not needed at 800 km.

**DESIGN DECISION 3 – DSP compensation configuration**

**I recommend a fully digital, DCF-free coherent receiver with the pipeline ADC/front-end correction → frequency-domain CD equalizer → timing recovery → 2×2 MIMO (CMA/RDE/DD-LMS, 21 taps) → frequency-offset estimation → blind-phase-search carrier recovery → SD-FEC, with DBP left out of the baseline but kept as a software-ready option, because** the CD (about 3.4 ns spread), PMD (about 2.8 ps) and laser phase noise are all linear and fully recoverable digitally; the ordering follows signal
dependency (static, large effects first; phase-blind adaptive stages before phase recovery); the resulting pre-FEC BER of about $1.3\times10^{-3}$ leaves a 3.6 dB margin at 800 km; and DBP would add 30–60 % DSP power for only about 0.5 dB, so it is only justified for links beyond about 2000 km.

**Stage 5 – Integrated WDM Transmission Strategy (15 marks)**

The integrated link delivers 6.4 Tb/s at 4 b/s/Hz with 3.6 dB margin at 800 km, at an estimated 135 pJ/bit including amplifiers; its main risks are the transmitter loss budget, Raman pump safety and laser/bias drift, all of which have defined mitigations.

### 5.1 Integrated architecture

Each channel has a silicon-photonic transmitter PIC (ITLA laser, 1×2 MMI polarization splitter, two nested IQ Mach-Zehnder modulators, polarization rotator and beam combiner) and a coherent receiver PIC (polarization-diversity 90° hybrids built from 2×4 MMIs, four germanium photodiodes, TIAs). Thirty-two channels are multiplexed in a 32-port AWG (50 GHz, Gaussian passband, 5 dB loss, ≥ 25 dB adjacent-channel isolation), amplified by a booster EDFA, sent through 10 hybrid Raman/EDFA spans with an optical isolator at each amplifier input and circulator-based Raman pump injection, pre-amplified, demultiplexed by a second AWG and received by 32 coherent receivers. Isolators (Faraday rotator, 35 dB isolation) stop back-reflections from destabilizing the laser, EDFAs and Raman gain, and the polarization rotator/beam combiner map the two modulated polarizations onto orthogonal fibre modes.

### 5.2 Transmitter loss budget and launch power (MMI/AWG impact)

| Stage | Loss or gain (dB) | Power per channel (dBm) |
|---|---|---|
| ITLA output laser | start | 14.0 |
| Fibre-to-chip coupling | −1.0 | 13.0 |
| 1×2 MMI splitter into X/Y polarizations (3.0 dB split + 0.3 dB excess) | −3.3 | 9.7 |
| 1×2 MMI splitter into I/Q branches | −3.3 | 6.4 |
| MZM insertion loss (3 mm silicon arm) | −5.0 | 1.4 |
| I/Q recombination MMI | −3.3 | −1.9 |
| 16QAM average-power backoff (drive at 0.35 Vπ, PAPR) | −4.0 | −5.9 |
| Polarization rotator/beam combiner | −1.5 | −7.4 |
| Output coupling to fibre | −1.5 | −8.9 |
| 32:1 AWG multiplexer | −5.0 | −13.9 (total 1.2 dBm) |
| Booster EDFA gain needed for −1.5 dBm launch | +12.4 | −1.5 (total 13.5 dBm) |

Every 1 dB of MMI or AWG excess loss costs 1 dB of booster gain (the booster has about 3.5 dB headroom to its 17 dBm saturation) and, because the booster NF is 5 dB, input OSNR into span 1 is 58 −
13.9 − 5 ≈ 39 dB, which is 10 dB better than the span-limited 28 dB, so transmitter-side losses up to about 5 dB extra are tolerable. A loss above that would degrade the first-span OSNR and the launch-power flatness.

### 5.3 Inter-dependencies

- **Format order, OSNR and amplifier:** each step from 16QAM to 64QAM needs about 5.7 dB more SNR, which would require roughly 4× more noise-free power (or about a quarter of the reach); this is why 16QAM sits at the knee and why the Raman stage (about 5 dB NF gain) is needed.
- **Format order and DSP:** 16QAM needs ENOB of at least 5.5 bit, a 32-angle blind phase search, a DD-LMS stage and about 15–20 % more DSP power than QPSK; the SD-FEC overhead (25 %) lifts the baud rate from 25 to 31.25 GBd.
- **Launch power and nonlinearity:** raising per-channel power by 1 dB above the optimum raises NLI by 3 dB, so the amplifier output limit and the modulator drive are coupled to the OSNR through the optimum launch power.
- **Modulator drive and OSNR:** a modulator with higher $V\pi$ forces a smaller drive and lower launched power, or a higher-swing, higher-power driver; thin-film LN or SOH would relieve this.
- **Channel spacing and filtering:** the AWG passband (about 37 GHz at −3 dB) must contain the 34.4 GHz signal; a narrower spacing adds crosstalk and filtering penalties.

### 5.4 Global performance estimate

| Parameter | Value |
|---|---|
| Aggregate throughput | 6.4 Tb/s net (8 Tb/s line) |
| Net spectral efficiency | 4 b/s/Hz |
| Total SNR at optimum launch, 10 spans, hybrid | 16.3 dB (theoretical threshold 12.7 dB) |
| Margin over pre-FEC threshold | 3.6 dB |
| Q-factor at optimum launch | about 9.6 dB (threshold 6.25 dB) |
| Pre-FEC BER | about $1.3\times10^{-3}$ (threshold $2\times10^{-2}$) |
| Power per bit (approximate estimate) | transceiver about 20 W per channel (about 100 pJ/bit); amplifiers about 7 W per channel over 11 sites (about 35 pJ/bit); total about 135 pJ/bit |
| Footprint | one Tx PIC and one Rx PIC per channel, each of order 5 × 10 mm; 32 channels as pluggable modules or four 8-channel line cards |

### 5.5 Trade-offs between reach, channel density and dynamic range

| Option | Channels in C-band | Net SE | Reach at 1 dB margin | Comment |
|---|---|---|---|---|
| 75 GHz spacing, DP-QPSK 62.5 GBd | 58 | 2.7 b/s/Hz | > 5000 km | Long-haul, lower capacity |
| **50 GHz spacing, DP-16QAM 31.25 GBd** | 88 slots | 4.0 | about 3200 km (hybrid) | Balanced |
| Option | Channels in C-band | Net SE | Reach at 1 dB margin | Comment |
| **(selected)** | (32 used) | b/s/Hz | | |
| 37.5 GHz flex grid, DP-16QAM 31.25 GBd | 117 | 5.3 b/s/Hz | about 2000 km (extra 1–2 dB crosstalk and filtering penalty) | Needs flex-grid WSS, tighter laser control |
| 50 GHz, DP-64QAM 20.8 GBd | 88 | 5.3 (with 37.5 GHz slot) | not feasible with a 17.5 dB transceiver SNR (needs 18.4 dB before any margin) | Would need a better transceiver (ENOB, driver linearity); reach figures in this table are approximate scaling estimates from the GN model |

The link is occupying 32 of about 88 available 50 GHz slots (36 %), so the system can be expanded with no change of technology; the Q-versus-launch-power curve in Stage 6 is flat within about 1 dB of its optimum, which sets the practical launch-power tolerance.

### 5.5 Critical failure points, scalability and standards

| Risk | Effect | Mitigation |
|---|---|---|
| Laser wavelength drift above ±2.5 GHz | Crosstalk and filter penalty | Wavelength locker, thermal control |
| MZM bias drift | Constellation distortion, 1–2 dB penalty | Automatic bias controller with dither tone |
| Raman pump failure or fibre cut | Loss of 10 dB span gain; high-power pump exposure hazard | Automatic pump shutdown, return-loss ≥ 40 dB, laser-safety interlock |

| | | |
|---|---|---|
| EDFA transients on channel add/drop | Power surges and burst errors | Fast gain clamping, per-channel power equalization |
| PDL accumulation | Statistical SNR loss of 0.5–1 dB | Component specification, MIMO equalizer |
| Filter concatenation in ROADMs | Narrowing of the 34.4 GHz signal | Keep to ≤ 4 WSS passes or use 37.5 GHz-wide filters |

Standards fit: the 50 GHz channel plan and the frequency anchor follow ITU-T G.694.1 and the multichannel application-code approach of G.698; the host-side electrical interface (2 × 100 Gb/s PAM4 lanes at 106.25 GBd) follows IEEE 802.3ck and OIF-CEI-112G; coherent DWDM with SD-FEC follows the architecture of IEEE 802.3ct 100GBASE-ZR and OIF 400ZR/ZR+. A 200 Gb/s DP-16QAM 31.25 GBd line side is not itself a ratified standard PHY, so the line side must be treated as a vendor-specific profile (as in OpenZR+) and interoperability must be validated by test.

**DESIGN DECISION 4 – Integrated WDM strategy**

**I recommend adopting the integrated strategy (silicon-photonic DP-16QAM transmitters, AWG multiplexing, hybrid Raman/EDFA spans, coherent DSP receivers with SD-FEC) for network deployment on 50 GHz grid up to about 1000 km, because** the cross-stage analysis shows 6.4 Tb/s at 4 b/s/Hz with 3.6 dB margin after 800 km, the transmitter loss budget is covered by a booster with 3.5 dB headroom, the DSP removes all linear impairments, the estimated 135 pJ/bit is within the range of deployed coherent pluggables, and the main risks (bias drift, laser drift, pump safety, transients) have standard, available mitigations.

**Stage 6 – Verification and Final System Design (10 marks)**

Every target in the specification is met in the analytical model, with a 3.6 dB SNR margin and BER of about $1.3\times10^{-3}$ against the $2\times10^{-2}$ pre-FEC limit; the table below lists the evidence for each line.

### 6.1 Verification table

| Requirement / performance parameter | Specified / target value | Achieved / evidence value |
|---|---|---|
| Aggregate WDM capacity | 6.4 Tb/s (32 ch × 200 Gb/s) | 6.4 Tb/s net, 8.0 Tb/s line rate (32 × 250 Gb/s with 25 % FEC) |
| Spectral efficiency and modulation choice | 4 b/s/Hz, DP-16QAM | 4.0 b/s/Hz net on 50 GHz grid; DP-16QAM at 31.25 GBd, 34.4 GHz occupied (RRC $\beta = 0.1$) |
| Modulator bandwidth and footprint | ≥ 25 GHz, compact PIC footprint | Silicon dual-pol IQ MZM, 3 dB bandwidth about 30–35 GHz (needs ≥ 22 GHz = 0.7 $R_s$), 3 mm arms, about 5 × 10 mm Tx PIC |
| Modulator drive voltage | Within SiGe driver capability | $V\pi$ about 6 V, drive 3–4 V peak-to-peak (0.35 $V\pi$ per arm) |
| Amplifier gain and gain flatness | 18 dB span gain, flat within ±0.5–1 dB | 10 dB Raman + 8 dB EDFA = 18 dB; ±0.5 dB after GFF; ±0.3 dB with WSS every 4 spans |
| Amplifier output power | 32 ch at 0 dBm assumed (15.05 dBm, 32 mW) | 13.5 dBm (22.6 mW) at −1.5 dBm per channel in the design; EDFA Psat 17 dBm |
| OSNR (10 spans, 0.1 nm) | ≥ 16.7 dB theory, plus margin | 28.3 dB ASE-only; 26.5 dB with NLI at the optimum launch power (SNR 22.5 dB + 3.98 dB) |

| | | |
|---|---|---|
| Pre-FEC BER / Q-factor | BER ≤ $2\times10^{-2}$, Q ≥ 6.25 dB | BER about $1.3\times10^{-3}$, Q about 9.6 dB including the 17.5 dB transceiver SNR |
| Margin over FEC | ≥ 3 dB at end of life | 3.6 dB (SNR 16.3 dB vs 12.7 dB) |
| Requirement / performance parameter | Specified / target value | Achieved / evidence value |
| threshold | | |
| DSP impairment mitigation | Full CD / PMD and nonlinear tolerance | CD 13 440 ps/nm removed by 421-tap-equivalent FDE; DGD 2.8 ps tracked by 21-tap MIMO; $\Delta\nu \cdot T_s$ = $6.4\times10^{-6}$ handled by BPS; nonlinearity handled by launch-power optimization (DBP optional, +0.3–0.5 dB) |
| Standards and feasibility | ITU-T G.698 and IEEE 802.3ck compliance | 50 GHz ITU-T grid; 802.3ck electrical host interface; coherent DWDM architecture as in 802.3ct / OIF 400ZR; the 200G DP-16QAM line side is vendor-specific and needs interoperability testing |

### 6.2 Block-level architecture

```
TRANSMITTER (x32, silicon-photonic PIC per channel)
  ITLA laser (f_k, <=100 kHz) --> 1x2 MMI (X/Y split)
      --> [X pol: IQ-MZM (I,Q arms, push-pull)] ---+
      --> [Y pol: IQ-MZM (I,Q arms, push-pull)] ---+--> polarization rotator + beam
combiner
  DAC 64 GS/s + RRC shaping + SiGe driver (3-4 Vpp) ---^                |
                                                                        v
                       32-port AWG multiplexer (50 GHz, 5 dB) --> isolator --> booster
EDFA (+12.4 dB)
                                                                        |
           x10 SPANS:  80 km SMF-28 <-- backward Raman pumps (2 x 1425/1455 nm via
circulator/WDM, ~0.9 W)
                       --> isolator --> GFF-EDFA (8 dB, NF 5 dB) --> [WSS gain
equalizer every 4th span]
                                                                        |
                                                              pre-amp EDFA --> 32-port
AWG demultiplexer
                                                                        |
RECEIVER (x32): LO laser (<=100 kHz) + polarization-diversity 90-degree hybrid (2x4
MMI) + 4 balanced Ge PDs + TIAs
   --> ADC 62.5 GS/s --> DSP: IQ correction --> CD FDE --> timing recovery --> 2x2
MIMO (CMA/RDE/DD-LMS)
   --> frequency-offset est. --> blind phase search --> demap --> SD-FEC (25 % OH) --
> 200 Gb/s client (802.3ck lanes)
```

### 6.3 System design parameter table

| Parameter | Value |
|---|---|
| Carrier frequencies / wavelengths | 32 carriers, 192.50 THz (1557.36 nm) to 194.05 THz (1544.91 nm), anchored to 193.1 THz |
| Parameter | Value |
| Grid spacing | 50 GHz (ITU-T G.694.1) |
| Laser type and linewidth | ITLA, ≤ 100 kHz (Tx) and ≤ 100 kHz (LO); output 14 dBm |
| Format, symbol rate, line rate | DP-16QAM, 31.25 GBd, 250 Gb/s (200 Gb/s net) |
| Pulse shaping | Root-raised-cosine, $\beta = 0.1$ |
| DAC / ADC | 64 GS/s / 62.5 GS/s, ENOB ≈ 5.5 bit |
| Modulator | Silicon carrier-depletion IQ-MZM, $V\pi \approx 6$ V ($V\pi L = 1.8$ V·cm, L = 3 mm), IL 5 dB, bandwidth ≥ 30 GHz |
| Transceiver back-to-back SNR | 17.5 dB (assumed from DAC/ADC ENOB, driver and receiver noise) |
| Multiplexers | 32-port AWG, 5 dB loss, ≥ 25 dB isolation; MMI 0.3 dB excess loss each |
| Raman amplifier | Backward, 2 pumps (1425, 1455 nm), about 0.9 W total, 10 dB on-off gain, effective span NF about 0 dB |
| EDFA | 8 dB gain in span, 17 dBm saturation, NF 5 dB, GFF ±0.5 dB |
| Fibre | SMF-28: $\alpha = 0.2$ dB/km, $D = 16.8$ ps/(nm·km), dispersion slope 0.056 ps/(nm²·km), $A_{eoo} = 80$ µm², $n_2 = 2.6\times10^{-20}$ m²/W, $\gamma = 1.3$ W$^{-1}$km$^{-1}$, PMD 0.1 ps/√km |
| Span / total length | 80 km, span loss 18 dB; 10 spans = 800 km |
| Launch power | −1.5 dBm per channel (13.5 dBm total) |
| DSP filter taps | CD: 1024-point FFT FDE (421 taps equivalent); MIMO 2×2, 4 × 21 taps at 2 sps; BPS 32 test phases |
| FEC | SD-FEC, 25 % overhead, pre-FEC BER threshold $2\times10^{-2}$ |

### 6.4 Simulation setup and expected results (OptiSystem / OptSim / MATLAB)

The schematic to build, component by component, follows the block diagram above: PRBS generator (2 independent sequences per polarization, $2^{15}-1$) → 16-QAM symbol mapper (4 bits/symbol) → RRC filter ($\beta = 0.1$, 31.25 GBd) → DAC → CW laser at each carrier (linewidth 100 kHz, 14 dBm) → dual-polarization IQ Mach-Zehnder modulator ($V\pi$ 6 V, bias at null) → 32-channel WDM multiplexer → loop control with 10 × [SMF-28 80 km ($\alpha$, D, $\gamma$, PMD set as above) → Raman (backward, two pumps) → EDFA with GFF] → WDM demultiplexer → coherent receiver (LO, 90° hybrid, photodiodes, TIA) → ADC → DSP chain (CD, timing, MIMO, FOE, CPE) → BER analyzer. Run with the nonlinear Schrödinger equation solver enabled and 32 channels active; simulate all 32 channels (or 7 channels with periodic boundary) because XPM depends on the neighbours.

Expected outputs to capture as screenshots:

- **Optical spectrum at the transmitter output:** 32 equal peaks 50 GHz apart over 1544.9–1557.4 nm, flat within about ±0.5 dB, each lobe about 34 GHz wide.
- **Optical spectrum after 10 spans:** same 32 channels with ASE floor about 26–28 dB below the signal in 0.1 nm.
- **Eye diagram (per I or Q tributary after the DSP, before decision):** 4-level signal, three eyes; with SNR 16.3 dB the relative error vector magnitude is $10^{-16.3/20} \approx 15$ %, so eyes remain open at about 4.5 σ spacing.
- **Constellation:** 16 clusters in a 4×4 grid, tight after CPE; before CD compensation a ring-shaped smear; before CPE rotated clusters.
- **Q-factor and BER versus launch power:** the table below.

| Launch power (dBm/ch) | EDFA-only SNR (dB) | EDFA-only BER | EDFA-only Q (dB) | Hybrid SNR (dB) | Hybrid BER | Hybrid Q (dB) |
|---|---|---|---|---|---|---|
| −6 | 13.0 | $1.7\times10^{-2}$ | 6.5 | 15.5 | $2.9\times10^{-3}$ | 8.8 |
| −4 | 14.1 | $8.5\times10^{-3}$ | 7.6 | 16.1 | $1.7\times10^{-3}$ | 9.3 |
| −3 | 14.6 | $6.2\times10^{-3}$ | 8.0 | 16.2 | $1.4\times10^{-3}$ | 9.5 |
| −2 | 14.9 | $4.7\times10^{-3}$ | 8.3 | 16.3 | $1.3\times10^{-3}$ | 9.6 |
| **−1.7 (hybrid optimum)** | 15.0 | $4.4\times10^{-3}$ | 8.4 | **16.3** | **$1.3\times10^{-3}$** | **9.6** |
| −1 | 15.2 | $3.9\times10^{-3}$ | 8.5 | 16.3 | $1.3\times10^{-3}$ | 9.6 |
| **0 (EDFA optimum)** | **15.3** | **$3.6\times10^{-3}$** | **8.6** | 16.1 | $1.6\times10^{-3}$ | 9.4 |
| 1 | 15.2 | $3.9\times10^{-3}$ | 8.5 | 15.8 | $2.1\times10^{-3}$ | 9.1 |
| 2 | 14.8 | $5.2\times10^{-3}$ | 8.2 | 15.3 | $3.5\times10^{-3}$ | 8.6 |
| 3 | 14.2 | $8.1\times10^{-3}$ | 7.6 | 14.5 | $6.4\times10^{-3}$ | 7.9 |

The curve shows the classic bell: SNR rises with power while ASE dominates, peaks where ASE and NLI balance, and falls at 3 dB per 1 dB of extra power once NLI dominates. Both links stay inside the $2\times10^{-2}$ limit over the whole range; EDFA-only at −6 dBm is only just inside it ($1.7\times10^{-2}$). These values come from the GN-model script below (10 spans, 17.5 dB transceiver SNR) and are not OptiSystem output; the OptiSystem run should reproduce them within about 1 dB, with the exact numbers depending on the DSP and FEC models.

```
import numpy as np
h,c,lam=6.626e-34,3e8,1550e-9; hv=h*c/lam
Rs,df,N=31.25e9,50e9,32; Btot=N*df
alpha=0.2/4.343/1e3; Lspan=80e3; gamma=1.3e-3
beta2=-16.8e-6*lam**2/(2*np.pi*c); Leff=(1-np.exp(-alpha*Lspan))/alpha
```

```
eta=(8/27)*gamma**2*(1/df)**3*Leff**2*np.arcsinh((np.pi**2/2)*abs(beta2)/alpha*Btot**
2)/(np.pi*abs(beta2)/alpha)*Rs
def snr_db(Pdbm,spans,NF_dB,loss_dB=18,trx_db=17.5,eps=0.05):
    P=1e-3*10**(Pdbm/10); G=10**(loss_dB/10); NF=10**(NF_dB/10)
    ase=NF*hv*G*Rs*spans; nli=eta*P**3*spans**(1+eps)
    s=P/(ase+nli); return -10*np.log10(1/s+10**(-trx_db/10))
print(snr_db(-1.7,10,0.0), snr_db(0,10,5.0))
```

This model uses the incoherent GN-model closed form for the nonlinear term (with a coherence exponent of 0.05) and, for the hybrid case, an effective span noise figure of 0 dB.

**Final Engineering Decision**

**YES – deploy the designed 32 × 200 Gb/s DP-16QAM DWDM system for spans up to about 10 × 80 km (800 km), with a pilot validation phase first.**

**Modulation format efficiency and OSNR sensitivity.** DP-16QAM at 31.25 GBd is the only candidate that fits 200 Gb/s into a 50 GHz slot (4 b/s/Hz, versus 2 b/s/Hz for 106 GBd PAM4 and 2.7 b/s/Hz for DP-QPSK). Its price is a 12.7 dB SNR (16.7 dB OSNR) requirement, 6.5 dB above QPSK, and the design delivers 16.3 dB total SNR, a 3.6 dB margin that also covers aging and PDL.

**Photonic modulator integration, bandwidth and drive voltage.** The silicon dual-pol IQ-MZM has about 30–35 GHz bandwidth (requirement 22 GHz = 0.7 of the baud rate) and a footprint of a few mm²; the Vπ of about 6 V needs a 3–4 V peak-to-peak SiGe driver, which is available. The 4–6 dB modulator loss and the three MMI splitting stages are absorbed by a booster with 3.5 dB headroom. The micro-ring was rejected for 80 pm/K thermal drift and a non-linear transfer function.

**Amplification gain profile, noise accumulation and power budget.** The hybrid span provides 18 dB gain (10 dB Raman + 8 dB EDFA), 13.5 dBm total output power for 32 channels, a flatness of ±0.5 dB with GFF and ±0.3 dB with periodic WSS equalization, and an OSNR of 28.3 dB (ASE only) after 10 spans, which is
3.3 dB better than an EDFA-only link (25 dB). The reach at 1 dB margin is about 3200 km versus 1600 km.

**DSP effectiveness against linear and nonlinear impairments.** CD (13 440 ps/nm, about 105 symbols of spread), DGD (2.8 ps), frequency offset and laser phase noise are fully compensated by the FDE, MIMO and BPS stages. Nonlinear impairments are controlled by choosing the optimum launch power and the dispersive SMF-28 (which suppresses FWM and averages XPM). DBP would add only about 0.3–0.5 dB for a 30–60 % increase in DSP power, so it is left out at this reach.

**Power, scalability and cost-effectiveness.** The estimated total of about 135 pJ/bit is in line with deployed coherent pluggables, the silicon PIC allows wafer-scale cost, and only 32 of 88 available 50 GHz slots are used, so capacity can grow to about 17.6 Tb/s on the same fibre and amplifiers without a technology change.

**Conditions attached to the YES.**

- Validate the numbers in OptiSystem/OptSim or on a hardware testbed, because the figures in this report come from a GN-model estimate, not from the full simulator; a deviation larger than about 1 dB in margin should trigger the redesign path below.
- Confirm the 17.5 dB transceiver SNR and the 5 dB modulator loss on real PICs.

- Verify interoperability separately, because the 200G DP-16QAM 31.25 GBd line side is not a ratified standard PHY.
- Implement the failure-point mitigations in Stage 5 (bias control, wavelength locking, Raman pump shutdown, EDFA fast gain clamping).

**Redesign path if validation fails (the NO branch).** If measured margin falls below 2 dB: (1) lower the order to DP-8QAM (about 41.7 GBd for 250 Gb/s line rate, 45.9 GHz occupied, so it still fits the 50 GHz slot and keeps 4 b/s/Hz net) or use probabilistically shaped 16QAM, either of which needs roughly 1.5–3 dB less SNR; (2) enable single-channel DBP (1 step per span, +0.3–0.5 dB); (3) shorten spans to 60 km (about 3 dB less span loss, +3 dB OSNR); or (4) move to thin-film LN modulators (Vπ about 2 V, loss about 2 dB) to raise launch OSNR and cut driver power. Predicted effect of (1) plus (3): about 4.5–6 dB extra margin (1.5–3 dB from the format change plus about 3 dB from shorter spans), with the 200 Gb/s per channel rate kept.

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